

# Enhancing toxicity prediction for natural products in herbal medicine and dietary supplements: Integrating (Q)STR models and *in vitro* assays

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## ABSTRACT

New approach methods (NAMs) are required to predict human toxicity effectively, particularly due to limitations in conducting *in vivo* studies. While NAMs have been established for various industries, such as cosmetics, pesticides, and drugs, their applications in natural products (NPs) are lacking. NPs' complexity (multiple ingredients and structural differences from synthetic compounds) complicates NAM development. In this study, we devised NAMs for NPs using (quantitative) structure-toxicity relationship (Q)STR models and *in vitro* assays.

Validation involved testing each method with single compounds isolated from NPs. A linear regression model was developed for (Q)STR prediction ( $R^2$  on test set: 0.52), with an applicability domain analysis demonstrating its reliability across NPs. This model was applied to predict the LD<sub>50</sub> range of species, aiding in the development of herbal medicine and dietary supplements. *In vitro* screening employed three reporter cell lines (AP-1, P53, and Nrf2), with Tox scores derived by integrating *in silico* and *in vitro* data.

Nimbolide exhibited the highest Tox score, with experimental studies corroborating the accuracy and reliability of the predictions made via Tox score analysis. The findings of the study align well with the purpose, as the suggested NAMs, utilizing (Q)STR models and *in vitro* assays, provide a Tox score to efficiently prioritize NPs for herbal medicine and dietary supplements.

## 1. Introduction

Natural products (NPs) have historically contributed to human health and continue to be utilized in herbal medicine and dietary supplements for their therapeutic properties (Wachtel-Galor and Benzie, 2011). Despite their perceived safety, reports of negative health impacts, including organ toxicities, cytotoxicity, and carcinogenicity, have surfaced (Bommakanti et al., 2023). The limitations of *in vivo* studies, such as ethical concerns and species-specific inaccuracies, have prompted the adoption of new approach methods (NAMs) for toxicity prediction (Jitäreanu et al., 2023). However, the effectiveness of NAMs at predicting NPs' toxicity requires validation. NPs are complex mixtures, making it challenging to evaluate NAM performance dependent on

variations resulting from extraction methods (Zhang et al., 2018) and seasonal and geographical variations (Li and Zidorn, 2022). Given that the inherent toxicity of NPs is dependent on their individual compounds (Woo et al., 2012), it is crucial to initially validate NAM performance for isolated NP molecules before extending their applicability to NP mixtures to explore potential synergetic effects. In this study, the term "NP" refers specifically to marker components or biologically active components, not extracts or mixtures of components.

*In silico* models and *in vitro* assays have emerged as NAMs for toxicity evaluation (Stucki et al., 2022). One widely used *in silico* approach is the (quantitative) structure-toxicity relationship (Q)STR model, which predicts molecule toxicity based on molecular structures alone. The reliability of (Q)STR models hinges on the chemical space of the training

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data (Weaver and Gleeson, 2008). However, the majority of (Q)STR models were developed using synthetic compounds, and some of the NPs extracted from plants were outside the model's applicability domain (AD) due to differences in chemical space. (Stratton et al., 2015). While (Q)STR models developed on synthetic compounds have been occasionally employed to predict the toxicity of NPs (Shin et al., 2023), it is imperative to develop models tailored to the chemical space of NPs for reliable outcomes (Seo et al., 2020).

The integrative approach often combines (Q)STR models and *in vitro* assays for toxicity screening. There are two methods for integrating *in silico* and *in vitro* data. One approach involves integrating assay results as features into the model along with structural information (Xu et al., 2020; Ye et al., 2022). However, the efficacy of adding extra assay features to enhance prediction accuracy is uncertain, as some models achieved similar or better accuracy with structure features alone. Another approach involves using *in silico* prediction and *in vitro* assay results independently (Chen et al., 2014; Jia et al., 2022). While machine learning models can be considered black-box models, making it difficult to explain prediction results, the latter approach provides transparent predictions by juxtaposing *in silico* predictions with *in vitro* data.

In this study, we independently integrated (Q)STR modeling and *in vitro* assays to predict the toxicity of NPs in herbal medicine and dietary supplements. The (Q)STR model was developed using NPs labeled with lethal dose to 50 % of test subjects ( $LD_{50}$ ) data from mice (*Mus musculus*) sourced from the NPASS database (Training set: 106, Test set: 27). Four machine learning algorithms were employed, using 2D descriptors and molecular fingerprints. The final model selection was based on prediction accuracy (Test set RMSE: 0.95 and  $R^2$ : 0.52), model interpretability, and AD of the model. *In vitro* assays, using lentiviral vectors to target activator protein 1 (AP-1), tumor protein P53 (P53), and nuclear factor erythroid-2-related factor 2 (Nrf2) to HepG2 cell lines, were integrated with (Q)STR and used for NP toxicity prioritization.

## 2. Materials and methods

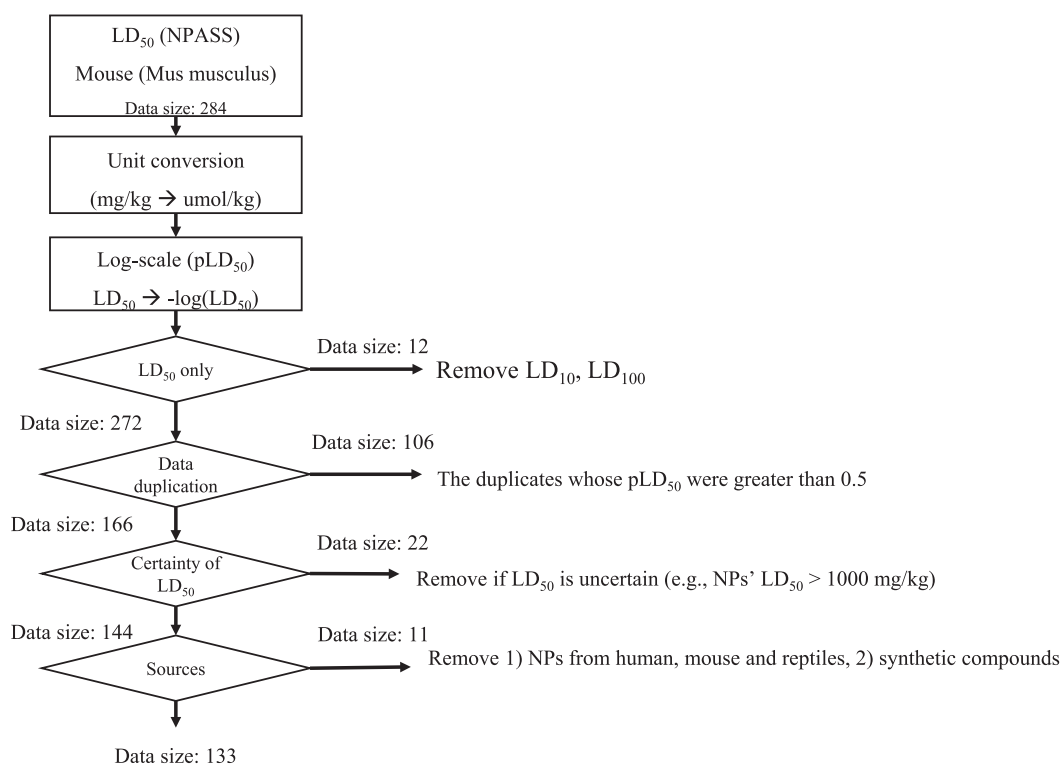
### 2.1. *In silico* model

#### 2.1.1. Natural products data collection and preprocessing

Data were sourced from the NPASS database (<https://bidd.gro.up/NPASS/>, accessed September 2023) (Zhao et al., 2023), which provides labeled NPs with activity and toxicity information included. To construct a (Q)STR model for predicting acute toxicity, NPs labeled with *Mus musculus*  $LD_{50}$  data were selected and preprocessed. First, unit of  $LD_{50}$  values were converted into  $\mu\text{mol/kg}$ , and then  $LD_{50}$  values were log-scaled with minus log base 10. Then, data with different labels such as  $LD_{10}$  and  $LD_{100}$  were excluded. Duplicated NPs were eliminated if the difference between the maximum and minimum  $pLD_{50}$  values exceeded 0.5. NPs with uncertain  $LD_{50}$  values (e.g.,  $LD_{50} > 1000 \text{ mg/kg}$ ) were also excluded. As this study focused on NPs derived from plants commonly used in herbal medicine or dietary supplements, NPs derived from mammals and reptiles were omitted (Fig. 1).

#### 2.1.2. Model development and validation

(Q)STR models were developed using the curated data described. The dataset was randomly divided into a training set and a test set, with 80 % (106 NPs) allocated to the training set and 20 % to the test set (27 NPs). For feature preparation, 2D descriptors were calculated using Mordred (Moriwaki et al., 2018) and quantum mechanical (QM) descriptors using MOPAC (<http://openmopac.net/>). Feature selection was conducted to find the optimum number of descriptors for the model. Four different algorithms—linear regression (LR), support vector regression (SVR), random forest (RF), and fully connected neural network (FNN)—were employed for model development. Scikit-learn was used for data splitting, feature selection, and training machine learning models. The best model was selected based on its prediction accuracy on the test set, and additional assessment of AD ensured the reliability of prediction results for NPs derived from plants potentially



**Fig. 1.** Data collection and preprocessing.  $LD_{50}$  data were collected from NPASS database.  $pLD_{50}$  was obtained after the unit was converted into  $\mu\text{mol/kg}$ . Since majority data were  $LD_{50}$ ,  $LD_{10}$  and  $LD_{100}$  were removed. Duplicated data were removed if discrepancies in  $pLD_{50}$  were over 0.5. NPs with ambiguous  $LD_{50}$  value were also removed from dataset. Finally, NPs from human, mouse and reptiles were removed.

used in herbal medicine and dietary supplements.

## 2.2. *In vitro* assays

Human hepatoma HepG2 cells, obtained from the American Type Culture Collection (ATCC), were used to establish cellular stress-sensing cell lines *via* lentiviral transduction with HepG2-GFP-AP1, HepG2-GFP-P53, and HepG2-GFP-Nrf2 (Park et al., 2024). These cell lines were cultured in Dulbecco's Modified Eagle's Medium (DMEM) (Lonza, Wokingham, UK) supplemented with 10 % fetal bovine serum (FBS) (Gibco) and 1 % penicillin/streptomycin (P/S) (Lonza, Wokingham, UK) at 37 °C with 5 % CO<sub>2</sub>. Cells were seeded at a density of 30,000 cells/well in a 96-well plate and then incubated for 24 h under the same conditions. Following exposure to compounds, cell staining was performed with 10 mg/ml Hoechst 33342 (Thermo Fisher, #H3570) and 2 mM Calcein-AM (Invitrogen, #C3100MP) and then analyzed using a high-content imaging system (HCS, Molecular Devices, CA, USA). Images were captured at 10× magnification using an ImageXpress® Micro XLS (Molecular Devices, CA, USA) with a Plan Fluor lens. DAPI filter cube confirmed Hoechst 33342, FITC confirmed GFP expression, and Texas Red confirmed Calcein-AM. Images were analyzed using MetaXpress 6 software (Molecular Devices, CA, USA), employing modules for nucleus counting and live/dead assessment. GFP-positive cells were enumerated, and their area and intensity were recorded.

### 2.2.1. Conversion of *in vitro* results into scores

*In vitro* assay results were converted into scores (0, 1, and 2) in order to combine *in vitro* results with *in silico* predictions. Each assay measures intensity of AP1, p53, and nrf2 with cytotoxicity as well (Fig. S1 ~ 3). Intensity and cytotoxicity were graded as strong (2), mild (1), and no effect (0). Strong intensity is greater than 200 % against control at the highest concentration. Mild intensity shows increase of intensity as concentration increases but less than 200 % at the highest concentration. Otherwise, intensity score was set to zero. Strong cytotoxicity shows cell viability lower than 50 % at the highest concentration. Mild cytotoxicity shows decrease of cell viability as concentration increases but higher than 50 % at the highest concentration. Otherwise, cytotoxicity score was set to zero. Therefore, each compound was scored by intensity and toxicity from the three stress-sensing cell lines. The highest intensity and cytotoxicity scores are six, respectively; thus, the total highest score is twelve and the lowest is zero.

### 2.2.2. Integration of *in silico* prediction and *in vitro* scores

The (Q)STR model predicted pLD<sub>50</sub> values, which are on a log scale, whereas the *in vitro* assay provided intensity and cytotoxicity scores of 0, 1, or 2 for three reporter cell lines. To balance the contribution of *in vitro* and *in silico* data in the final score calculation, we summed the intensity and cytotoxicity scores from the *in vitro* assays and divided the sum by three, reflecting the data obtained from three different cell lines.

$$\text{in vitro score} = \frac{\sum (\text{intensity score}_{\text{cell line}} + \text{cytotoxicity score}_{\text{cell line}})}{3 \text{ (the number of reporter cell lines)}}$$

Therefore, the maximum achievable score in the *in vitro* assay is four, while the minimum is zero. The intensity score was determined by summing the intensity scores for AP-1, P53, and Nrf2, and the cytotoxicity score was the sum of the cytotoxicity scores for each reporter cell line. The resulting *in vitro* score was then added to the *in silico* value to calculate the final score, termed the Tox score.

$$\text{Tox score} = \text{pLD}_{50(\text{in silico})} + \text{in vitro score}$$

## 2.3. Case study

As a case study, we focused on predicting the toxicity of NPs using NAMs. The selected NPs were sourced from plants used in herbal medicine or dietary supplements; however, their toxicity profiles were not

well defined. Hence, we applied (Q)STR models and *in vitro* assays to predict and analyze the toxicity of the selected NPs. Since toxicity profiles are not clear, we try to evaluate the final prediction through available information from research articles.

## 3. Results

### 3.1. (Q)STR model prediction

#### 3.1.1. Model data analysis

The data were analyzed to assess the chemical space and pLD<sub>50</sub> distribution. Fig. 2A illustrates the chemical space, represented by the molecular weight (MW) and partition coefficient (logP) of NPs alongside their pLD<sub>50</sub> distribution. NPs with higher MW and logP tended to exhibit relatively high pLD<sub>50</sub> values. In Fig. 2B, pLD<sub>50</sub> values displayed substantial overlap between the training and test sets, confirming the reliability of the model's validation. Given the objective of developing a (Q)STR model capable of accurately predicting NP toxicity for dietary supplements and herbal medicine, NPs derived from species such as poisonous reptiles and mammals were excluded, as they are unlikely candidates for use in dietary supplementary or herbal medicine development. Following the removal of such species from the dataset, species distribution was examined (Fig. 2C), revealing that the majority of NPs originated from plants.

#### 3.1.2. Feature selection and model performance

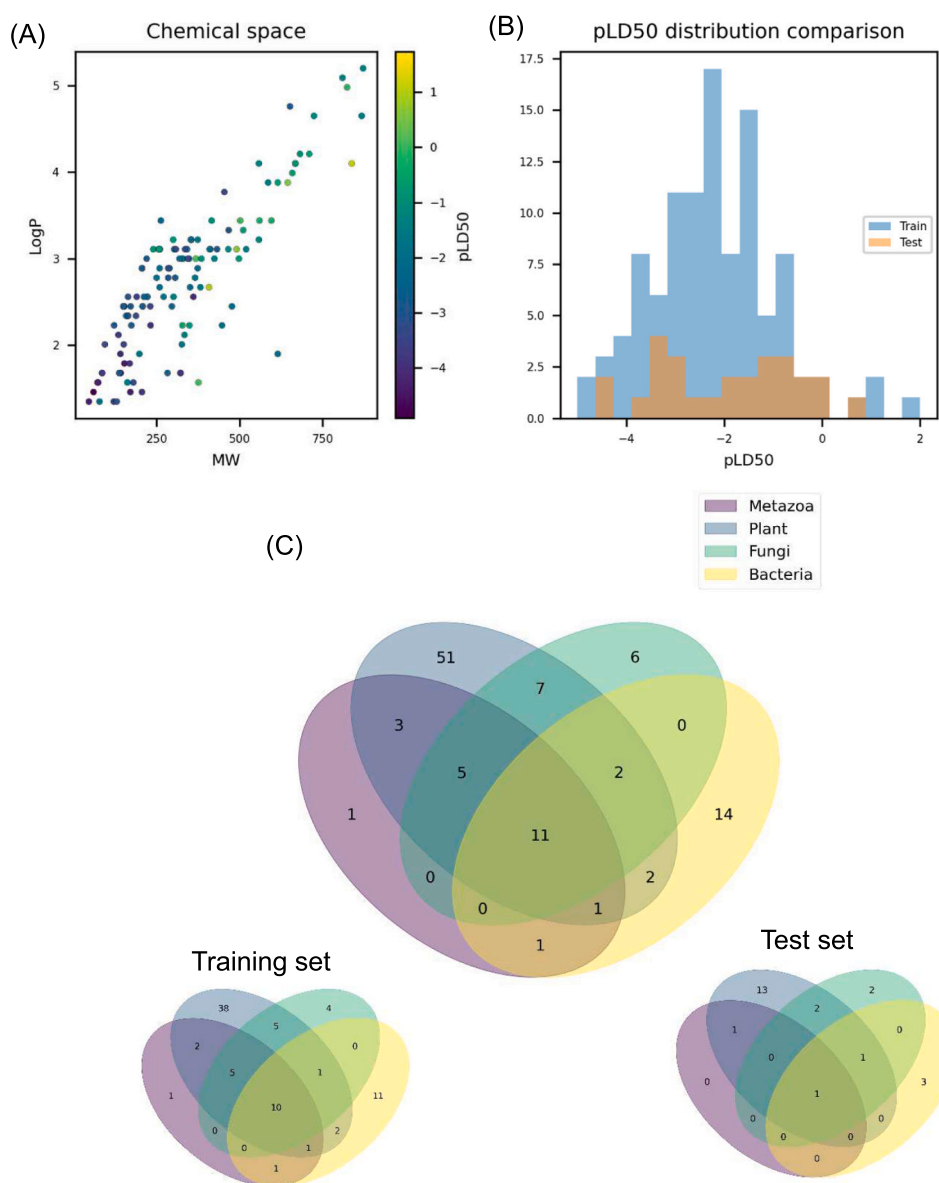
(Q)STR models were developed using a training set comprising 106 NPs and validated with test set of 27 NPs. 2D and QM descriptors were combined within the dataset for model development. Forward selection was employed for four distinct machine learning algorithms, *i.e.*, LR, SVR, RF, and FNN. LR and SVR models showed similar patterns of improvement in prediction accuracy (Fig. 3A and B). RF and FNN models achieved high prediction accuracy in the training set but not in test set (Fig. 3C and D). Due to the limited size of the training set, models that achieved the highest prediction accuracy on the test set with the fewest number of descriptors were prioritized. Notably, Fig. 3 illustrates that despite the application of complex machine learning algorithms and an increase in the number of descriptors, prediction accuracy did not improve. LR models with only two descriptors displayed superior prediction accuracy compared to other algorithms (Table 1); hence, LR models were chosen as the final model for predicting NPs' toxicity.

#### 3.1.3. Applicability domain analysis

The LR model demonstrated commendable prediction accuracy on both the training and test sets, with AD analysis confirming its reliability (Fig. 4). As the model's intended application is on NPs beyond the training set, AD analysis was expanded to encompass the entire NPASS database. Notably, a substantial number of NPs fell outside the models' AD (Fig. 5A). After curation of NPs based on the source species of NPs, all the selected NPs were within AD (Fig. 5B). Based on AD analysis results, pLD<sub>50</sub> values were predicted for the selected NPs (Table S1). Inherent species toxicity was a focal point in NPs' toxicity prediction. Therefore, pLD<sub>50</sub> values for each species were predicted. In this context, pLD<sub>50</sub> values were predicted for all NPs from a species only if they fell within the model's AD; species with any NPs outside the AD were excluded (Fig. 5C, Table S2). While the (Q)STR model can't predict synergic effects of mixtures, it can be applied to predict the range of pLD<sub>50</sub> values based on NPs from a species, which could serve as supplementary information to identify possible molecules contributing to the inherent toxicity of species.

## 3.2. *In vitro* assays

NPs selected for the case study were subjected to testing using reporter cell lines. NPs were scored based on the intensity and the viability, categorized as strong (2), mild (1), or no effect (0). In cases



**Fig. 2.** Data analysis after preprocessing. (A) chemical space was examined with molecular weight and partition coefficient (logP). Dots were colored according to pLD<sub>50</sub> values. Color is brighter as pLD<sub>50</sub> increases and darker as it decreases. (B) pLD<sub>50</sub> distribution is examined between training set and test set. (C) Species distribution was examined after preprocessing. Majority natural compounds were from plant, which is a dominant source for herbal medicine and dietary supplement.

where decreased intensity resulted from increased cell death, the intensity score was assigned based on the dose-response pattern before significant cell death occurred. An intensity score of strong was designated if the intensity increased by over 200 % compared to the control, while a mild intensity score was given when intensity increased with concentration but did not exceed 200 % at the highest concentration. A strong cytotoxicity score was given when the lowest cell viability fell below 50 %, whereas a mild cytotoxicity score was assigned if cell viability decreased with concentration but remained above 50 %. The *in vitro* scores are outlined in Table 2.

### 3.2.1. AP-1 sensing cell line (intensity and cell viability)

Based on GFP intensity, six NPs showed increase in AP-1 expression (Decursinol angelate, Nimbolide, 5,7-demethoxyflavone, Hypericin, Podophyllotoxin, and Xanthatin). Notably, Nimbolide and Podophyllotoxin displayed the strongest AP-1 expression. All six NPs led to a decrease in the survival rate of HepG2 cells. Conversely, three NPs

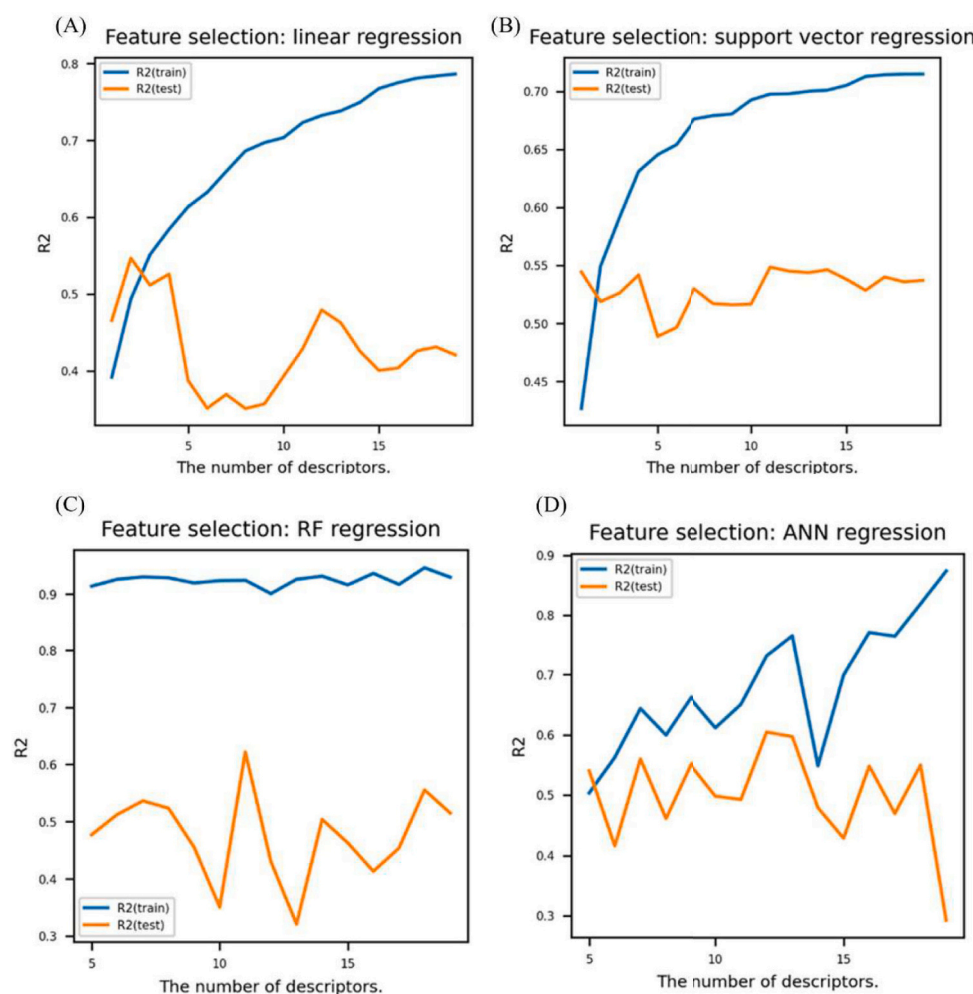
(Piperine, Silybin, and Biochanin A) showed a decrease in cell survival rate without an increase in AP-1 intensity. Nimbolide demonstrated robust AP-1 expression and significant cytotoxicity (Fig. S1).

### 3.2.2. P53 sensing cell line

P53 intensity and cytotoxicity increased with three NPs (Nimbolide, Podophyllotoxin, and Xanthatin). Although Hypericin showed an increase in P53 intensity, cell viability remained unaffected. Conversely, eight NPs showed a decrease in cell viability without a corresponding increase in P53 intensity (Decursinol angelate, Calycosin, 5,7-Dimethoxyflavone, Piperine, Baicalein, Silybin, Biochanin A, and Hydroquinone). Notably, Xanthatin demonstrated strong P53 expression and significant reduction in cell viability (Fig. S2).

### 3.2.3. Nrf2 sensing cell line

Nrf2 intensity and cytotoxicity increased with seven NPs (Decursinol angelate, Calycosin, Nimbolide, 5,7-Dimethoxyflavone, Baicalein,



**Fig. 3.** Forward selection was conducted on features comprised of 2D and QM descriptors. Linear regression model (A) and support vector regression (B) showed similar performance on training and test set. Random forest (C) and neural network (D) achieved highest prediction accuracy in training set as the number of descriptors were increased. However, prediction accuracy in test set wasn't significantly improved.

**Table 1**

Prediction accuracy of the linear regression model with the different number of descriptors.

| The number of descriptors | Selected descriptor* | R <sup>2</sup> |      | RMSE  |      | Relative RMSE |         | Spearman's corr. |      |
|---------------------------|----------------------|----------------|------|-------|------|---------------|---------|------------------|------|
|                           |                      | Train          | Test | Train | Test | Train         | Test    | Train            | Test |
| 1                         | CIC0                 | 0.38           | 0.48 | 1.01  | 0.99 | 15 %          | 14.86 % | 0.68             | 0.71 |
| 2                         | Estate_VSA4          | 0.45           | 0.52 | 0.95  | 0.95 | 14.28 %       | 14.26 % | 0.72             | 0.76 |
| 3                         | ATSC8c               | 0.51           | 0.41 | 0.9   | 1.05 | 13.47 %       | 15.72 % | 0.74             | 0.69 |
| 4                         | ATSC6m               | 0.55           | 0.31 | 0.87  | 1.14 | 13.00 %       | 17.09 % | 0.77             | 0.55 |
| 5                         | AATSC3p              | 0.57           | 0.31 | 0.84  | 1.13 | 12.63 %       | 17.04 % | 0.79             | 0.57 |
| 6                         | SdNH                 | 0.59           | 0.42 | 0.82  | 1.04 | 12.31 %       | 15.64 % | 0.81             | 0.66 |

\* Descriptors were selected through forward selection. Therefore, each descriptor in a row was added to the descriptors previously selected.

Hydroquinone, and Xanthatin). No NPs showed strong Nrf2 expression. Two NPs (Aloin and Hypericin) increased Nrf2 expression without a reduction in cell viability, while four NPs (Podophyllotoxin, Piperine, Silybin, and Biochanin A) decreased cell viability without a corresponding increase in Nrf2 intensity (Fig. S3).

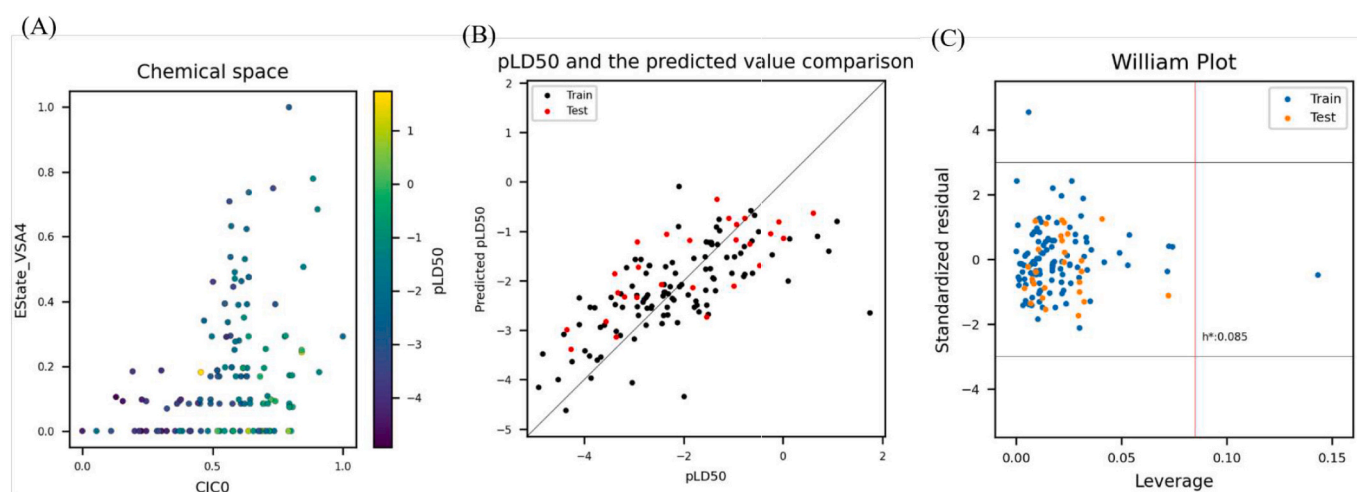
### 3.3. Case study: Tox score calculation

Each NP was scored based on both *in vitro* and *in silico* data, which were then integrated to calculate the Tox score. Among the screened NPs, *in vitro* score ranged from 3.33 to 0, with the highest value assigned to Xanthatin and the second highest to Nimbolide. The *in silico* values

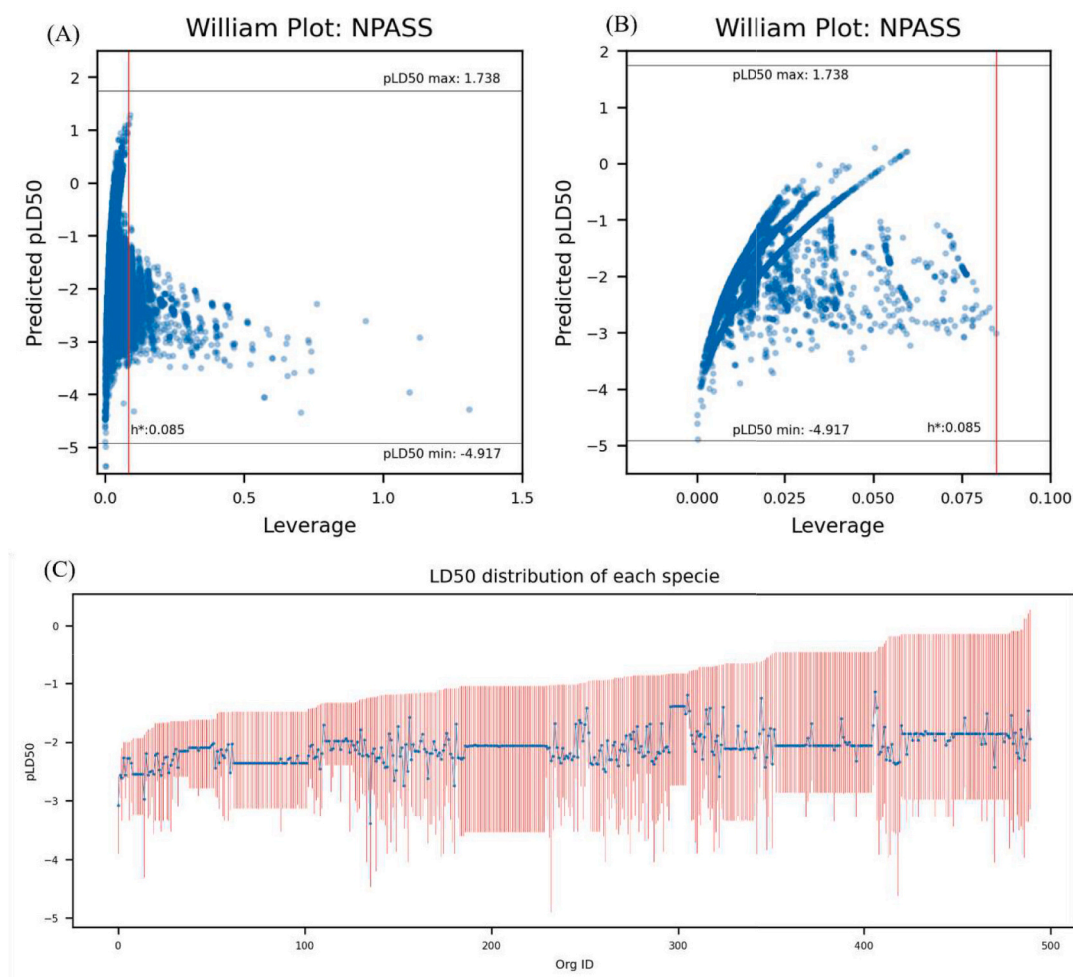
ranged from −0.4 to −3.16, with the highest value attributed to Ginsenoside Rg1 and the second highest to Ecdysterone. In the final score, Nimbolide emerged as the most toxic NP with Tox score of 1.47, while Xanthatin ranked as the second most toxic NP with a score of 1.063. Although the *in vitro* score alone was higher for Xanthatin than for Nimbolide, the *in silico* value favored Nimbolide. Consequently, Nimbolide had the highest Tox score (Table 2).

## 4. Discussion

This study developed NAMs based on (Q)STR and *in vitro* assays to predict toxicity of NPs for herbal medicine and dietary supplements. The



**Fig. 4.** Linear regression model analysis. (A) Chemical space was re-examined with the selected descriptors. (B) Parity plot showed that model achieved good prediction accuracy for train and test set. (C) Applicability was examined through William plot, and all the data was in AD except two NPs.



**Fig. 5.** Applicability domain analysis and pLD<sub>50</sub> prediction over NPs in NPASS database.

(Q)STR model was developed using a dataset curated from NPASS database, with the target endpoint being the LD<sub>50</sub> value of *Mus musculus*. Four machine learning algorithms were employed in model development, and the linear regression model achieved comparable prediction

accuracy to other algorithms in test set. Therefore, the linear regression model was selected for *in silico* prediction due to its lower risk of over-fitting and transparent prediction result analysis. In order to examine prediction reliability of the model, AD analysis was conducted.

**Table 2**  
Incorporation of (Q)STR and *in vitro* data for NPs' toxicity prediction.

| Sources                          | Ingredients                  | (Q)STR                          | AP-1               |                    | P53                |                    | Nrf2               |                    | Tox score    |
|----------------------------------|------------------------------|---------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------|
|                                  |                              | pLD <sub>50</sub> <sup>1)</sup> | Int. <sup>2)</sup> | Tox. <sup>3)</sup> | Int. <sup>2)</sup> | Tox. <sup>3)</sup> | Int. <sup>2)</sup> | Tox. <sup>3)</sup> |              |
| Achyranthes bidentata Blume      | Ecdysterone                  | -0.72                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -0.72        |
| Aloe arborescence                | Aloin                        | -1.43                           | 0                  | 0                  | 0                  | 0                  | 1                  | 0                  | -1.1         |
| Angelica gigas                   | Decursinol angelate          | -1.86                           | 1                  | 2                  | 0                  | 2                  | 1                  | 2                  | <b>0.807</b> |
| Astragalus membranaceus          | Calycosin                    | -2.01                           | 0                  | 0                  | 0                  | 1                  | 1                  | 1                  | -1.01        |
| Azadirachta Indica A. Juss.      | Nimbolide                    | -1.53                           | 2                  | 2                  | 1                  | 2                  | 1                  | 1                  | <b>1.47</b>  |
| Boesenbergia rotunda             | 5,7-Dimethoxyflavone         | -2.02                           | 1                  | 2                  | 0                  | 2                  | 1                  | 2                  | <b>0.647</b> |
| Camellia sinensis, Thea sinensis | (-)-Epigallocatechin gallate | -2.31                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -2.31        |
| Cynanchum wilfordii              | 4-Hydroxyacetophenone        | -2.77                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -2.77        |
| Ephendra sinica                  | (1R,2S)-(-)-Ephedrine        | -2.34                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -2.34        |
| Gynostemma pentaphyllum Makino   | Gypenoside VIII              | -0.92                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -0.92        |
| Hypericum perforatum L.          | Hypericin                    | -1.62                           | 1                  | 1                  | 1                  | 0                  | 1                  | 0                  | -0.29        |
| Linum usitatissimum L.           | Podophyllotoxin              | -1.71                           | 2                  | 1                  | 1                  | 1                  | 0                  | 1                  | 0.29         |
| Panax ginseng                    | Ginsenoside Rg1              | -0.4                            | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -0.4         |
| Piper longum L.                  | Piperine                     | -2.89                           | 0                  | 1                  | 0                  | 2                  | 0                  | 1                  | -1.56        |
| Polygala tenuifolia              | Tenuifolin                   | -0.88                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -0.88        |
| Polygala tenuifolia              | Polygalaxanthone III         | -1.44                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -1.44        |
| Scutellaria baicalensis          | Baicalin                     | -2.31                           | 0                  | 0                  | 0                  | 1                  | 1                  | 1                  | -1.31        |
| Silybum marianum                 | Silybin                      | -1.47                           | 0                  | 1                  | 0                  | 2                  | 0                  | 2                  | 0.197        |
| Tagetes erecta                   | Lutein                       | -0.73                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -0.73        |
| Tribulus terrestris L.           | Solanine                     | -1.06                           | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -1.06        |
| Trifolium pratense L.            | Biochanin A                  | -2.17                           | 0                  | 1                  | 0                  | 1                  | 0                  | 1                  | -1.17        |
| Xanthium sibiricum               | Atractyloside                | -1.5                            | 0                  | 0                  | 0                  | 0                  | 0                  | 0                  | -1.5         |
| Xanthium sibiricum               | Hydroquinone                 | -3.16                           | 0                  | 0                  | 0                  | 1                  | 1                  | 1                  | -2.16        |
| Xanthium strumarium L.           | Xanthatin                    | -2.27                           | 1                  | 2                  | 2                  | 2                  | 1                  | 2                  | <b>1.063</b> |

\*Intensity and cell survival were graded with three different levels such as strong (2), mild (1), and no effect (0). Strong intensity is greater than 200 % against control at high concentration. Mild intensity shows increase of intensity as concentration increases but less than 200 % at high concentration. Strong cytotoxicity shows cell viability lower than 50 % at high concentration. Mild cytotoxicity shows decrease of cell viability as concentration increases but higher than 50 % at high concentration.

1) Unit of LD<sub>50</sub> is umol/kg/day.

2) Int. is intensity.

3) Tox. is cytotoxicity.

According to AD analysis result, entire NPs in NPASS can't be reliably predicted by the model developed in this study as large number of NPs in NPASS dataset was found outside the model's AD (Fig. 5A). The fact that the model was trained on mouse (*mus musculus*) LD<sub>50</sub> data limits its AD. Also, data set was further preprocessed by removing NPs isolated from human, mouse, and reptiles. A second AD analysis was performed using only NPs from species in the dataset (Fig. 2C). All the filtered NPs were placed within the AD of the model (Fig. 5B). This indicated that the source species information was significant in AD analysis. As researchers are also interested in predicting the source species' toxicity, the (Q)STR model was further applied to predict the LD<sub>50</sub> range of species based on the NPs with identified molecular structures (Fig. 5C). Although the model cannot predict synergetic effects between NPs in the species, it can help identify potential toxicants among the ingredients. The predicted LD<sub>50</sub> range of species in the NPASS database can be useful for selecting species in the development of herbal medicine and dietary supplements.

Three reporter cell lines (AP-1, P53, and Nrf2) were utilized in toxicity screening for NPs. AP-1 controls crucial cellular processes such as differentiation, proliferation, and apoptosis through regulation of gene expression in cytokines, growth factors, and stress (González-Rubio et al., 2016; Shaulian and Karin, 2022). P53 influences various cellular processes by regulating over 500 genes, including those involved in oxidative phosphorylation and reactive oxygen species (ROS) generation (Simabuco et al., 2018). Nrf2 regulates antioxidant genes, playing a critical role in protecting cells from oxidative stress (Ramachandran and Saraswathy, 2014). Therefore, employing three *in vitro* cell lines based on these mechanisms can serve as a primary screening method for

predicting NPs' toxicity. The Tox score suggested in this article integrates *in silico* and *in vitro* data. To balance the contribution of each method to toxicity prediction, the *in vitro* assay scores were averaged and summed with the *in silico* prediction.

According to the Tox score, Nimbolide exhibits the highest score. Nimbolide is a compound isolated from the leaves and flowers of *Azadirachta indica* (Pooladanda et al., 2019). In an acute toxicity study of Nimbolide, the LD<sub>50</sub> value was reported as 24 mg/kg (pLD<sub>50</sub>: -1.95) in adult male mice (Glinsukon et al., 1986). In the (Q)STR model, Nimbolide's predicted pLD<sub>50</sub> value is -1.53, corresponding to 15.805 mg/kg. *Azadirachta indica* leaf extract was reported to be cytotoxic, and Nimbolide was identified as the key compound for this cytotoxicity (Chitta et al., 2014). The molecular mechanism of Nimbolide's cytotoxicity suggests that it activates apoptosis by inhibiting the cytoprotective action (Sophia et al., 2018). Nimbolide demonstrated an increase in AP-1, P53, and Nrf2 expression along with cytotoxicity. Currently available toxicity information on Nimbolide aligns well with the (Q)STR prediction and *in vitro* results. Xanthatin was the second highest NP by the Tox score that using the results with the developed AP1, P53, and Nrf2 to HepG2 cell lines. Previous research has yielded reports on the toxicity of Xanthatin and the toxicity of *Xanthium strumarium* L., which is main components (Chen et al., 2023; Fan et al., 2019; Romero et al., 2014). The predicted pLD<sub>50</sub> for xanthatin was determined to be -2.27, corresponding to a dose of 45.864 mg/kg/day. Based on the reported cytotoxic results, Xanthatin was selected as a candidate compound for confirming toxicity results in AP1, P53, and Nrf2 cell lines. The results supported the basis for NAMs for predicting the toxicity of natural products containing multiple components, as indicated by a significant

Tox score in the developed (Q)STR model.

The objective of this study was to develop novel NAMs based on the (Q)STR model for predicting acute toxicity in mice and to elucidate the underlying mechanisms through *in vitro* assays. The reporter cell lines proposed in this study can provide initial mechanism-based toxicity screening of new materials based on natural ingredients (Park et al., 2024). This is obtained through the evaluation of the expression of GFP intensity as well as cytotoxicity in each reporter cell line. Therefore, through the integration of results from *in vitro* assays and (Q)STR models, the Tox scores provided by NAMs can effectively prioritize NPs for potential use in herbal medicines and dietary supplements.

In conclusion, further research on *in vitro* assays and the newly proposed NAMs will facilitate the development of drugs based on multi-component natural products and enable the creation of novel, complex predictive toxicity testing methods for multi-component systems, such as natural products.

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## CRediT authorship contribution statement

**Hyun Kil Shin:** Writing – original draft, Visualization, Investigation. **Se-Myo Park:** Writing – original draft, Investigation, Formal analysis. **Mi-Sun Choi:** Investigation, Data curation. **Jung-Hwa Oh:** Resources, Conceptualization. **Sang Kyum Kim:** Validation, Data curation. **Seok-joo Yoon:** Funding acquisition, Data curation. **Hae-Ryung Park:** Supervision, Methodology. **Hyoung-Yun Han:** Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no competing interests.

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## Data availability

Data will be made available on request.

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